FOUNDATION COURSE

START YOUR PREPARATION FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 12

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Recap *of previous lecture*

TRIGONOMETRY



Topics *to be covered*

"MATRICES"



Defn Matrix is the rectangular arrangement of $m \cdot n$ numbers

ie $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m \times n} = [a_{ij}]_{m \times n}$ where $1 \le i \le m$, $1 \le j \le n$

where i = Row & j = Column

m = Counting of No of Rows.

& n = Counting of No. of Columns.

→ Horizontal Lines are called Rows & Vertical lines are called Columns

→ If $m = n$ or No of Rows = No. of Columns then such Matrices are called Square Matrices

⊛ a_{ij} = elements of Matrix

(*) Row Mat → For this $m=1$, eg $A = \begin{bmatrix} 1 & -2 & 3 & 4 & 2 \end{bmatrix}_{1\times 5}$

$B = \begin{bmatrix} 2 & -1 & 4 \end{bmatrix}_{1\times 3}$

(*) Column Mat → For this $n=1$, eg $C = \begin{bmatrix} 2 \\ -1 \\ 4 \\ 2 \end{bmatrix}_{4\times 1}$, $D = \begin{bmatrix} 2 \\ -1 \end{bmatrix}_{2\times 1}$

(*) Null Mat → if all the elements in a Matrix are Zero then it is Null Mat

$O_{2\times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}_{2\times 2}$, $O_{2\times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{2\times 3}$, $O_{3\times 1} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}_{3\times 1}$

Types of Sq. Matrices ($m=n$) →

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} = [a_{ij}]_{4\times 4}$$

4×4

In any Matrix there are two diagonals shown in the diagram in which

PINK shaded Diagonal = Principal Diag or Main Diag

Here a_{ij} (for $i=j$) are called Diag elements.

Note:- ① for diag elements, $i=j$ $\forall i \& j$

② for upper diag elements, $i<j$ $\forall i \& j$

③ for lower " " , $i>j$ " "

④ for dff diag elements, $i \neq j$ " "

⑤ Corresponding elements are a_{ij} & a_{ji}

Upper Triangular Mat , Lower Triangular Mat , Diag Mat , Scalar Mat

U.T.M

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{bmatrix}$$

L.T.M

$$\begin{bmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & 0 & 0 & 0 \\ 0 & a_{22} & 0 & 0 \\ 0 & 0 & a_{33} & 0 \\ 0 & 0 & 0 & a_{44} \end{bmatrix}$$

$$\begin{bmatrix} k & 0 & 0 & 0 \\ 0 & k & 0 & 0 \\ 0 & 0 & k & 0 \\ 0 & 0 & 0 & k \end{bmatrix}$$

eg $\begin{bmatrix} 2 & -1 & 4 & 0 \\ 0 & -1 & 1 & 2 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix}$ UTM

eg $\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 2 & 0 & 0 & 0 \\ 0 & 1 & -4 & 0 & 0 \\ -1 & 0 & 2 & 0 & 0 \\ 0 & 1 & -1 & 2 & 4 \end{bmatrix}$ LTM

eg $\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$ D.Mat , eg $\begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix}$ S.Mat 3×3

eg $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ Id.Mat. 4×4

Diag Mat — $A=[a_{ij}]_{n\times n}$ s.t $a_{ij}=\begin{cases}0 & \text{when } i\neq j\\ \text{atleast one element must be Non zero} & \text{when } i=j\end{cases}$,

then A is called Diag Mat.

Identity Mat (unit Mat) $\longrightarrow A=[a_{ij}]_{n\times n}$ where $a_{ij}=\begin{cases}0, & i\neq j\\ 1, & i=j\end{cases}$

eg $I_2=\begin{bmatrix}1&0\\0&1\end{bmatrix}_{2\times 2}$, $I_3=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}_{3\times 3}$, $I_4=\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix}_{4\times 4}$ & so on....

Note ① $AI=IA=A$

② $AO=OA=O$,

Equal Matrices – A and B are said to be equal if

(1) they are of same order (2) Each element is also also same.

eg $A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ & $B=\begin{bmatrix} x & y \\ z & w \end{bmatrix}$ s.t $A=B$ then $a=x, b=y, c=z, d=w$

2×2 $\quad 2\times2$

eg $A=\begin{bmatrix} 2 & x & -3 \\ 4 & -1 & 0 \end{bmatrix}$ & $B=\begin{bmatrix} 2 & -1 & -3 \\ x+y & -1 & 0 \end{bmatrix}$ s.t $A=B$ then find x & $y=?$

2×3 $\quad 2\times3$

$\Rightarrow x=-1$ and $4=x+y$

$4=-1+y$

$y=5$

Matrix Addition & Subtraction –

If $A=\begin{bmatrix} a & b & c \\ c & d & f \end{bmatrix}$ then $A\pm B=\begin{bmatrix} a\pm\alpha & b\pm\beta & c\pm\gamma \\ c\pm x & d\pm y & f\pm z \end{bmatrix}$

& $B=\begin{bmatrix} \alpha & \beta & \gamma \\ x & y & z \end{bmatrix}$

eg $A=\begin{bmatrix}2 & -1\\ 1 & 4\\ 1 & 0\end{bmatrix}_{3\times2}$ & $B=\begin{bmatrix}0 & 4\\ -1 & 2\\ 1 & 3\end{bmatrix}_{3\times2}$

$A+B=\begin{bmatrix}2 & 3\\ 0 & 6\\ 2 & 3\end{bmatrix}=B+A$

$A-B=\begin{bmatrix}2 & -5\\ 2 & 2\\ 0 & -3\end{bmatrix}$ & $B-A=\begin{bmatrix}-2 & 5\\ -2 & -2\\ 0 & 3\end{bmatrix}$

(*) Matrix addition or subtraction is possible only when they are of same order
or Two Matrices are said to be conformable for addition if they are of same order

(*) Matrix addition follows Commutative Law as well as Associative Law
i.e $A+B=B+A$ (Commutative law)
& $(A+B)+C=A+(B+C)$ (Associative Law)

(*) Matrix Multiplication is Associative But Not Commutative (in general)
i.e $(AB)C=A(BC)$ But $AB\neq BA$

Multiplication of Matrices → process will be cleared in following Examples;

Let, $A_{m\times n}$ & $B_{n\times p}$ then $(AB)_{m\times p}$ for eg $A_{2\times 3}$ & $B_{3\times 4}$ then → AB = Defined, BA = N.Def.

is N.Cond^{n} for Multiplication →

No. of Columns in 1st Mat = No of Rows in 2nd Mat while

$\because A_{2\times 3}\ B_{3\times 4}$ = Defined

$B_{3\times 4}\ A_{2\times 3}$ = N.D. ??

eg $A=\begin{bmatrix}2 & -1 & 4\\ 0 & 1 & 2\end{bmatrix}_{2\times 3}$

& $B=\begin{bmatrix}1 & -1\\ 2 & 0\\ 3 & 4\end{bmatrix}_{3\times 3}$

Then AB & BA = ?

Sol :→ $AB=\begin{bmatrix}2 & -1 & 4\\ 0 & 1 & 2\end{bmatrix}_{2\times 3}\begin{bmatrix}1 & -1\\ 2 & 0\\ 3 & 4\end{bmatrix}_{3\times 2}=\begin{bmatrix}a_{11} & a_{12}\\ a_{21} & a_{22}\end{bmatrix}=\begin{bmatrix}12 & 14\\ 8 & 8\end{bmatrix}$

where $a_{11}=(2)(1)+(-1)(2)+(4)(3)=12$

$a_{12}=(2)(-1)+(-1)(0)+(4)(4)=14$

$a_{21}=(0)(1)+(1)(2)+(2)(3)=8$

(ii) $B=\begin{bmatrix}1 & -1\\ 2 & 0\\ 3 & 4\end{bmatrix}$, $A=\begin{bmatrix}2 & -1 & 4\\ 0 & 1 & 2\end{bmatrix}$

$$BA=\begin{bmatrix}1 & -1\\ 2 & 0\\ 3 & 4\end{bmatrix}_{3\times 2}\begin{bmatrix}2 & -1 & 4\\ 0 & 1 & 2\end{bmatrix}_{2\times 3}=\begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{bmatrix}$$

$$=\begin{bmatrix}2 & -2 & 2\\ 4 & -2 & 8\\ 6 & 1 & 10\end{bmatrix}_{3\times 3}$$

$a_{13}=(1)(4)+(-1)(2)$

$=4-2=2$

$BA=\begin{bmatrix}1\\ 2\\ -3\end{bmatrix}_{3\times 1}\begin{bmatrix}-2 & 1 & 4\end{bmatrix}_{1\times 3}=\begin{bmatrix}a_{11}\end{bmatrix}_{1\times 1}=\begin{bmatrix}-12\end{bmatrix}_{1\times 1}$

$a_{11}=(-2)(1)+(1)(2)+(4)(-3)$

$=-2+2-12=-12$

$AB=\begin{bmatrix}1\\ 2\\ 3\end{bmatrix}_{3\times 1}\begin{bmatrix}-2 & 1 & 4\end{bmatrix}_{1\times 3}=\begin{bmatrix}-2 & 1 & 4\\ -4 & 2 & 8\\ -6 & 3 & 12\end{bmatrix}_{3\times 3}$

Q $A=\begin{bmatrix}1\\ 2\\ -3\end{bmatrix}_{3\times 1}$, $B=\begin{bmatrix}-2 & 1 & 4\end{bmatrix}_{1\times 3}$ then

$AB=?=(AB)_{3\times 3}$

$BA=?=(BA)_{1\times 1}$

i.e. $B_{1\times 3}A_{3\times 1}=(BA)_{1\times 1}$

⊛ Scalar Multiple

Let $A=\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}_{2\times 3}$

& k be any scalar (constant)

then $kA = k\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} = \begin{bmatrix} ka & kb & kc \\ kd & ke & kf \end{bmatrix}$

eg: $A=\begin{bmatrix} 2 & -1 & 4 \\ 1 & 0 & 2 \\ 1 & 1 & 4 \end{bmatrix}_{3\times 3}$ then $-3A = ? = \begin{bmatrix} -6 & 3 & -12 \\ -3 & 0 & -6 \\ -3 & -3 & -12 \end{bmatrix}$

⊛ Transpose of Matrix

if $A=[a_{ij}]_{m\times n}$ then $A^T=[a_{ji}]_{n\times m}$

i.e. By interchanging Rows & Columns, we can find Transpose.

for eg $A=\begin{bmatrix} 2 & -1 & 4 & 1 \\ 3 & 0 & 1 & 2 \end{bmatrix}_{2\times 4}$

then $A^T = ? = \begin{bmatrix} 2 & 3 \\ -1 & 0 \\ 4 & 1 \\ 1 & 2 \end{bmatrix}_{4\times 2}$

eg $A=\begin{bmatrix} 2 & -1 & 1 \\ 1 & 3 & 2 \\ 0 & 1 & -4 \end{bmatrix}_{3\times 3}$ then $A^T = ?$

$A^T=\begin{bmatrix} 2 & 1 & 0 \\ -1 & 3 & 1 \\ 1 & 2 & -4 \end{bmatrix}_{3\times 3}$

(1) $(A\pm B)^T = A^T \pm B^T$

(2) $(A^T)^T = A$

MINORS & COFACTORS :- (M_{ij}) (C_{ij})

$$A=\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$ then

$$M_{11}=\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}, \quad C_{11}=(-1)^{1+1}\cdot M_{11}$$

$$M_{12}=\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}, \quad C_{12}=(-1)^{1+2} M_{12}$$

$$M_{32}=\begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix}, \quad C_{32}=(-1)^{3+2} M_{32}$$

i.e. $\boxed{C_{ij}=(-1)^{i+j} M_{ij}}$

i.e. for $A_{3\times 3}$, $C_{ij}=\begin{bmatrix} +M_{11} & -M_{12} & +M_{13} \\ -M_{21} & +M_{22} & -M_{23} \\ +M_{31} & -M_{32} & +M_{33} \end{bmatrix}$

adjoint of Mat :- Let A is the given Mat

then $\boxed{Adj(A)=(Cof\,A)^T}$

Inverse of Mat :- if A is given Mat s.t $|A|\neq 0$

then, Inverse of $A=\dfrac{adj\,A}{|A|}$ or $\boxed{A^{-1}=\dfrac{adj\,A}{|A|}}$

Note: $A^{-1}=\dfrac{1}{A}$ wrong

Singular Mat :- if $|A|=0$ then A is called Singular.

Q $A=\begin{bmatrix}2 & -1 & 4\\ 0 & 1 & -3\\ -2 & 3 & 5\end{bmatrix}$,

Minor of $-2=\begin{vmatrix}-1 & 4\\ 1 & -3\end{vmatrix}$,

$M_{31}=(3)-(4)=-1 \Rightarrow C_{31}=+(-1)$

Minor of '2' $=\begin{vmatrix}1 & -3\\ 3 & 5\end{vmatrix}=(5)-(-9)=14=M_{11} \Rightarrow C_{11}=(-1)^{1+1}M_{11}=14$

" of $-1=\begin{vmatrix}0 & -3\\ -2 & 5\end{vmatrix}=(0)-(6)=-6=M_{12} \Rightarrow C_{12}=(-1)^{3}M_{12}=6$

" of $4=\begin{vmatrix}0 & 1\\ -2 & 3\end{vmatrix}=(0)-(-2)=2=M_{13} \Rightarrow C_{13}=(-1)^{4}M_{13}=2$

" " $0=\begin{vmatrix}-1 & 4\\ 3 & 5\end{vmatrix}=(-5)-(12)=-17=M_{21} \Rightarrow C_{21}=-(-17)=17$

" " $1=\begin{vmatrix}2 & 4\\ -2 & 5\end{vmatrix}=(10)-(-8)=18=M_{22} \Rightarrow C_{22}=+(18)$

" " $-3=\begin{vmatrix}2 & -1\\ -2 & 3\end{vmatrix}=(6)-(2)=8=M_{23} \Rightarrow C_{23}=-(8)$

Minor of $3=$

$M_{32}=\begin{vmatrix}2 & 4\\ 0 & -3\end{vmatrix}$

$=(-6)-(0)=-6$

$C_{32}=(-1)^{3+2}M_{32}=+6$

Minor of $5=$

$M_{33}=\begin{vmatrix}2 & -1\\ 0 & 1\end{vmatrix}=2$

$C_{33}=(-1)^{3+3}M_{33}=2$

$C_{ij}=\begin{bmatrix}14 & 6 & 2\\ 17 & 18 & -8\\ 1 & 6 & 2\end{bmatrix} \Rightarrow \text{adj } A=(C_{ij})^{T}$

(ii) Also Find $adjA$ & A^{-1} = ?

$$adjA = (\text{Cof } A)^T = \begin{bmatrix} 14 & 17 & -1 \\ 6 & 18 & 6 \\ 2 & -8 & 2 \end{bmatrix}$$

Now $|A| = \begin{vmatrix} 2 & -1 & 4 \\ 0 & 1 & -3 \\ -2 & 3 & 5 \end{vmatrix} =$

$= +2[5-(-9)] - 0 + (-2)[3-4]$

$= 28 + 2 = 30$

So $A^{-1} = \dfrac{adjA}{|A|} = \dfrac{1}{30}\begin{bmatrix} 14 & 17 & -1 \\ 6 & 18 & 6 \\ 2 & -8 & 2 \end{bmatrix}$

Shortcut of finding adj of $A_{2\times 2}$ →

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $adjA = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Q. Find the cof A for which $A=\begin{bmatrix}2 & -3\\ 4 & -5\end{bmatrix}$ and adj A as well as $A^{-1}=?$

Sol $C_{11}=(-1)^2 M_{11}=+(-5)=-5$

$C_{12}=(-1)^3 M_{12}=-(4)=-4$

$C_{21}=(-1)^3 M_{21}=-(-3)=+3$

$C_{22}=(-1)^4 M_{22}=+(2)=2$

$\therefore \text{cof}A=\begin{bmatrix}C_{11} & C_{12}\\ C_{21} & C_{22}\end{bmatrix}=\begin{bmatrix}-5 & -4\\ 3 & 2\end{bmatrix}$

$\text{adj}A=(\text{cof}A)^T=\begin{bmatrix}2 & 3\\ -4 & -5\end{bmatrix}$

Now $|A|=(-10)-(-12)=2$

$\therefore A^{-1}=\frac{\text{adj}A}{|A|}=\frac{1}{2}\begin{bmatrix}2 & 3\\ -4 & -5\end{bmatrix}$

Q. if $A=\begin{bmatrix}-5 & 7\\ -8 & 4\end{bmatrix}$ then adj A = ? & $A^{-1}=?$

Using Shortcut: $\text{adj}A=\begin{bmatrix}4 & -7\\ 8 & -5\end{bmatrix}$

(ii) $|A|=(-20)-(-56)=+36$

$\therefore A^{-1}=\frac{\text{adj}A}{|A|}=\frac{1}{36}\begin{bmatrix}4 & -7\\ 8 & -5\end{bmatrix}$

Reversal Law $\Rightarrow$ $(ABC)^T = C^T B^T A^T$

$adj(ABC) = (adj\,C)(adj\,B)(adj\,A)$

$(ABC)^{-1} = C^{-1} B^{-1} A^{-1}$

Cancellation Law in Matrix Algebra $\Rightarrow$

$$AB = AC \Rightarrow B = C$$

(only when A is Non Singular i.e when A^{-1} exist)

Proof: Let $|A| \neq 0$ i.e A is Non Singular

then $A^{-1}(AB) = A^{-1}(AC)$

$(A^{-1}A)B = (A^{-1}A)C \Rightarrow IB = IC \Rightarrow B = C$


Thank
YOU